# Ibtsam Ayyaz

**QA Automation Engineer | SQA Engineer | Software Test Engineer**

Lahore, Pakistan | +923014272979 | ibtsamayyaz01@gmail.com | https://www.linkedin.com/in/ibtsamayyaz/

## Professional Summary

QA Automation Engineer / SDET with 4+ years of experience in web, API, mobile, and hardware-integrated testing. Builds scalable Cypress and Playwright automation and AI-driven QA agents: built the Cypress-to-Playwright migration and mutation-testing agents, and contributed to a 14-agent suite covering test generation, PR review, triage, and release impact analysis. Grew regression coverage from an initial 3,000 Cypress tests to roughly 7,000, and cut full-suite runtime from 3-4 hours to about 30 minutes through Jenkins parallelization. Strong in CI/CD, SQL validation, and failure investigation.

## Technical Skills

**Automation & Languages:** Playwright, Cypress, JavaScript, TypeScript, Node.js

**API Testing:** Postman, REST APIs, OpenAPI / Contract Testing

**Database:** SQL Server, MySQL, T-SQL

**CI/CD & DevOps:** Jenkins, GitHub, AWS

**Mobile Testing:** Android testing, iOS testing, mobile testing (iOS), manual mobile testing

**Tools:** GitHub Copilot, Claude Code, MCP server development, AI-driven QA agents, JIRA, Xray, Confluence

**Testing Practices:** Agile methodologies, Test reporting, SDLC, STLC, Regression, Smoke, Integration, Exploratory, Functional Testing, Mutation testing, Agent-assisted testing

## Professional Experience

### QA Automation Engineer II | TCP Software

*Nov 2024 - Present*

- Designed and maintained scalable Cypress and Playwright automation frameworks using Page Object Model, inline scripting, and reusable JSON-driven execution.
- Managed regression and sanity suites, growing coverage from an initial 3,000 Cypress test cases to roughly 7,000 tests, with new tests added continuously.
- Built the Migration and Mutation Testing agents and contributed to 12 other AI-driven QA agents, for a 14-agent suite: a seven-agent pipeline the team still uses end to end to add new test coverage (Feature Explorer, Test Planner, Test Investigator, Test Generator, Test Executor, Test Analyzer, and Test Maintenance, in that order), plus five independent agents: PR Reviewer, Triage, AC Conformance, Release Impact Analyzer, and Exploratory Testing
- Built a guarded Cypress-to-Playwright migration agent, migrating about 500 tests per day and completing the suite migration in six days; migrated tests passed quality checks, including assertion-level mutation testing.
- Developed a mutation-testing agent that creates a separate mutated test for each assertion and verifies the test fails at the intended assertion because of the mutation.
- Parallelized Cypress Jenkins runs by splitting npm execution into independent groups and validating data, login, and test dependencies, reducing full-suite runtime from 3-4 hours to about 30 minutes.
- Built Node.js utilities and Jenkins retry workflows to read Xray failures, map tests to databases, restore databases, and rerun only failed tests with configurable retries; reduced a two-person regression recovery cycle from 1-1.5 days to about 4-5 hours.
- Personally built an MCP server that automates failed-test re-execution: given Xray IDs, it adds/removes .only on the relevant Cypress specs, determines required run inputs, executes specs in parallel terminals, and updates both a parent terminal and the JIRA ticket with live pass/fail status -- replacing a manual, one-by-one re-run process; also supports standalone lookups (spec names, database names, terminal commands by Xray ID)
- Contributed to agent-assisted PreQA feature/bug-fix testing and sprint/release certification: agents use repository code and JIRA ticket context to prepare test plans and risk reports, then execute tests; QA reviews failures, guides recovery, or performs blocked checks manually. Increased team capacity from 1-2 to 3-6 tickets per person, depending on complexity and available agent capacity.
- Introduced AI-assisted automation using GitHub Copilot and, currently, Claude Code, accelerating test development and reducing manual scripting effort
- Tested APIs and validated backend data with SQL across workforce and payroll systems; investigated flaky tests and reduced Cypress failures from 400+ to about 80 during the first year.
- Integrated automation into Jenkins CI/CD pipelines, reducing manual testing effort by about 20 hours per sprint.
- Performed manual Android and iOS testing of workforce-management workflows, including clock-in/out, geolocation-based attendance, leave and scheduling workflows, real-time data synchronization, and cross-device usability.
- Conducted hardware testing of biometric time-clock devices and validated device connectivity and user workflows.
- Reported sprint regression results (pass/fail counts via Xray) through a shared JIRA ticket, giving Product, Engineering, and QA leadership self-serve, real-time visibility into regression status.
- Contributed to team documentation, runbooks, and best-practice guides for the automation framework and testing processes.

### SQA Engineer | TECHVERX

*July 2021 - July 2024*

- Built and maintained Cypress-based automation framework using Page Object Model
- Implemented data-driven testing for financial trading workflows at LightPoint Financial Technologies
- Performed backend validation using Microsoft SQL Server to ensure financial data accuracy
- Conducted API testing using Postman, validating request/response flows and system behavior
- Executed regression, integration, functional, and exploratory testing in an Agile team environment, across roughly 39 release cycles over 3 years (2-week sprints, 2 sprints per release)
- Wrote test plans and logged defects in JIRA to track and communicate quality issues
- Collaborated closely with developers and product teams to debug issues, clarify requirements, and improve feature quality

## Education

**BS Computer Science** | COMSATS University Islamabad | 2017 - 2021